Math 340
Practice Final

You are allowed a calculator and one 8.5” x 11” sheet of handwritten notes (both sides). The actual exam will have 14 questions. The Table of Laplace Transforms will be provided on the actual final. The practice exam is longer so you have a better sense of the types of problems that may be asked. Note that I often include a question on the actual exam that is not similar to questions on the practice exam.

Problem 1

Find all solutions to $\frac{dy}{dx} = e^{x-y}$

Problem 2

Solve the initial value problem $\frac{dy}{dx} = \frac{3x-2y}{2x+4y}$, $y(0) = 2$.

Problem 3

Find all solutions to $\frac{dy}{dx} + \mathrm{y} = \frac{x}{y^2}$.

Problem 4

Use the improved Euler’s method with step size $h = 0.5$ to approximate $y(1.5)$ where

$\frac{dy}{dx} = x^2 - y^2$, $y(0)=1$.

Problem 5

A tank has 10 kg of salt dissolved in 200 liters of water. Water with a salt concentration of 30g/liter flows into the tank at a rate of 2 liters/min, and the (perfectly mixed) water in the tank leaves at the same rate. How much salt is in the tank after an hour?

Problem 6

Solve the initial value problem $y'' + 6y' + 8y = 0, \quad y(0) = 1, y'(0) = 1$.

Problem 7

Find the general solution to $y'' + 9y = \cos(3x)$.

Problem 8

Write $4\exp(i\pi/3)$ in rectangular form, $x + iy$.

Problem 9

A mass of 2kg is attached to a spring, causing it to stretch 0.05m (5 centimeters). What amount of damping is critical for the spring-mass system? Use $g = 9.8\text{m/sec}^2$.

Problem 10

Find the general solution to $x^2y'' + 4xy' + 2y = 0$.

Problem 11

Solve the initial value problem $x'' + 4x' + 5x = \sin(t^2)\,, x(0) = 0, x'(0) = 0$. Your answer will have an integral you can't simplify.

Problem 12

Solve the initial value problem $x'' + 2x' + 5x = \delta(t) + \delta(t-1), \quad x(0) = 0, x'(0) = 0$.

Problem 13

Find the a priori lower bound for the radius of convergence of the series solution to $(x^3 - 2x^2 - 3x - 20)y'' + xy' - 3y = 0$ about $x_0 = 0$. Hint: $x = 4$ is a singular point.

Problem 14

Solve the initial value problem $y'' + (2x + 1)y' + y = 0$, $y(0) = 1, y'(0) = 1$. Write the series out to the x^4 term.

Problem 15

Solve the initial value problem.

$$\frac{dx}{dt} = 4x + 4y, \;\; x(0) = 4$$
$$\frac{dy}{dt} = x + 4y, \;\; y(0) = 8$$

Problem 16

Solve the convolution equation for $x(t)$: $x(t) = \exp(-t) + \int_0^t \exp(t - \tau)\, x(\tau) d$

Problem 17

Find and classify the critical points (equilibria) for the system

$$\frac{dx}{dt} = x^2 - y^2$$
$$\frac{dy}{dt} = x^2 + 2y^2 - 3$$

Problem 18

Draw the curves $x' = 0$ and $y' = 0$ and sketch several (parametric) curves defined by the system

$$\frac{dx}{dt} = x^2 - y$$
$$\frac{dy}{dt} = x^2 + y^2 - 2$$

Problem 19

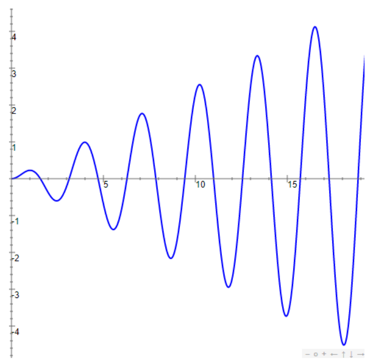
Triton orbits Neptune with a semi-major axis of about $3.549 \times 10^8 m$ and orbital period about $5.078 \times 10^5 sec$. What is the approximate mass of Neptune?
Use $G \approx 6.67 \times 10^{-11} m^3 kg^{-1} sec^{-2}$ for the gravitational constant.

Problem 20

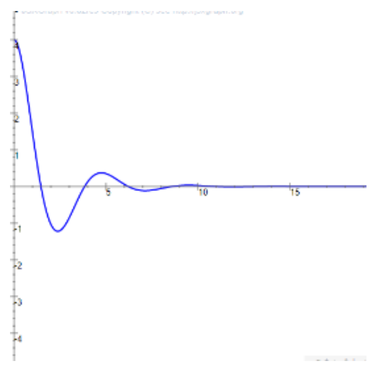
A rocket is sent from the Earth (at 1 A.U.) to Mars (at 1.5 A.U.) along a Hohmann transfer orbit. How long will the trip take?

Problem 21

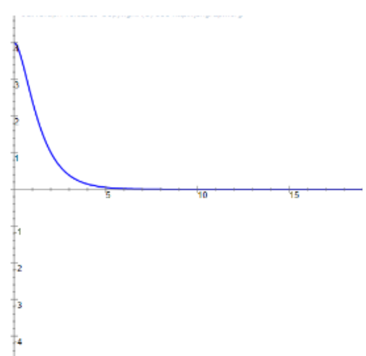
The following three equations each have a missing number. The three missing numbers are **1, 3, and 4**. Fill in the correct missing number for each equation in order to get the solution curve at the right of each equation. **You must fill in a different number for each equation**.



(a) $x'' + \_\_\_ x = \cos 2t$



(b) $x'' + \_\_\_ x' + 2x = 0$



(c) $x'' + \_\_\_ x' + 2x = 0$
